



Original research article

Analysis of Mathematical and Graphical Relationships Between Volume, Speed, and Density of Traffic with Greenshields and Underwood Models (Case Study: Ir. H. Juanda Street, Central Jakarta)

Reinhart Ignasius Nugroho^{a*}, Imam Hagni Puspito^a

^a Civil Engineering Department, Universitas Pancasila, 12640, Indonesia

ARTICLE INFO

Keywords:

Graphical
Greenshields
Mathematical
PKJI
Underwood

ABSTRACT

DKI Jakarta Province is one of the centers of transportation progress in Indonesia. One of the locations around Jakarta that is of great concern is the Juanda area, specifically on Ir. H. Juanda Street, Central Jakarta. The daily activity of motorized vehicle users on this road looks very congested on busy days such as Wednesday and Friday and quieter on Sunday. All vehicle activities on the road can be modeled to determine the relationship based on the traffic flow characteristics. The data obtained in the field, such as volume and speed, will be processed first with basic relationships using PKJI standard, then can be analyzed with mathematical and graphical relationships to present the relationship between volume, speed, and density of traffic on Wednesday, Friday, and Sunday on Ir. H. Juanda Street, Central Jakarta with Greenshields and Underwood models. The analysis results according to the relationship between volume, speed, and density of traffic with PKJI standard is suitable for Ir. H. Juanda Street on the Underwood model because it shows the strongest relationship with the r value of 0,940475 between speed and density to produce the equation $U_s = 46,22 \exp(-0,02 D)$, $Q = 55,84 U_s \ln(46,22/U_s)$, and $Q = 46,22 D \exp(-0,02 D)$. The Underwood model has the best accuracy on Sunday, with the r^2 value of 0,885001. The determination coefficient shows how well the regression prediction of the model is based on field data with an accuracy of 88,5%. The highest maximum volume was obtained using the Underwood model on Wednesday at 1.319,07296 SMP/hour.

1. Introduction

DKI Jakarta Province is one of the centers of transportation progress in Indonesia. The speed of urbanization experienced by the City of Jakarta has caused a lot of traffic jams at many points, especially in Central Jakarta City, which is the center of government and economy.

One of the locations around Jakarta that is of great concern is the Juanda area, specifically on Ir. H. Juanda Street, Central Jakarta. This area is one of the metropolitan areas with very wide transportation access, and its development is fast.

A metropolitan area must have a land transportation infrastructure that covers all parts of the road, including complementary buildings and equipment intended for traffic [1].

As part of the road, an urban road segment is a road segment with characteristics on one or both sides that are built permanently, continuously, and thoroughly along the corridor, has morning and evening peak traffic, traffic flow is dominated by passenger cars and motorcycles, percentage of trucks large and small, and there is a curb [2].

The capacity of urban roads must be separated into several

segments if the road characteristics change significantly. Changes in carriageway and shoulder width (up to 15%), road type, visibility, type of road alignment, and exits from urban or semi-urban areas, even if the geometric or other characteristics do not change. Urban road capacity analysis is only carried out for vertical alignment types that are flat or almost flat and horizontal alignment types that are straight or nearly straight [3].

Ir. H. Juanda Street was chosen as one of the widest roads between the intersections before entering the road. The condition of this road has not yet been disturbed by construction work such as the MRT construction in the Harmoni area, so it is considered feasible and effective for research.

The daily activity of motorized vehicle users on Ir. H. Juanda Street, Central Jakarta, looks congested on busy days such as Wednesday and Friday and quieter on Sunday. All vehicle activities on the road can be modeled to determine the relationship based on the traffic flow characteristics.

The characteristics of the traffic flow movement on a road section and the ability of the road segment to accommodate traffic flow need special attention. The following are three

*Corresponding author: Civil Engineering Department, Universitas Pancasila, 12640, Indonesia

E-mail address: 4218215025@univpancasila.ac.id (Reinhart Ignasius Nugroho)

doi: [10.21776/ub.rekayasasipil.2025.019.01.9](https://doi.org/10.21776/ub.rekayasasipil.2025.019.01.9)

Received: 13 October 2024; Revised: 23 December 2024 Accepted: 6 January 2025

E-ISSN: 2502-6348 © 2025 rekayasasipil@ub.ac.id. All rights reserved.

main parameters that affect the operational characteristics of traffic flow: volume (Q), speed (U_s), and density (D) [4].

The three main parameters called the basic relationship can apply variable substitution if the values of the two variables are known so that the other variables can be calculated using Eq. (1) [4], [5].

$$Q = U_s \cdot D \quad (1)$$

This basic relationship must be described using an analytical model to form traffic characteristics in mathematical equations and a simulation model to depict traffic forecasts in graphical lines.

The mathematical relationship between volume, speed, and density variables used is based on two models, namely based on field observations (linear regression approach) called the Greenshields model and based on boundary conditions called the Underwood model.

1. Greenshields Model

This model is the earliest to observe traffic behavior. Greenshields (1934) conducted a study on Ohio suburban roads, where traffic conditions met the requirements because they were uninterrupted and moved steadily (steady state condition) [6]. Greenshields found that the relationship between speed and density is linear, as shown in Eq. (2).

$$U_s = U_f - \left(\frac{U_f}{D_f} \right) \cdot D \quad (2)$$

The relationship between volume and density in Eq. (3) can be found by substituting Eq. (2) with Eq. (1).

$$Q = U_f \cdot D - \left(\frac{U_f}{D_f} \right) \cdot D^2 \quad (3)$$

This equation is the parabolic equation $Q = f(D)$ [7]. If $D = Q/U_s$, then based on Eq. (3), the volume and speed relationship is obtained in Eq. (4).

$$Q = D_j \cdot U_s - \left(\frac{D_j}{U_f} \right) \cdot U_s^2 \quad (4)$$

This equation is also a parabolic function between $Q = f(U_s)$. So, it can be concluded that if there is a linear relationship between speed and density, then the relationship between speed and volume and volume and density will function parabolic [4], [8]. The maximum volume (Q_m) can be calculated using Eq. (5) [9].

$$Q_m = \frac{D_j \cdot U_f}{4} \quad (5)$$

2. Underwood Model

This model is taken directly from the boundary conditions of points on the basic curve of volume, speed, and density [10]. Drew and Underwood show that the hypothesis of traffic volume is an exponential relationship between speed and density expressed in Eq. (6).

$$U_s = U_f \cdot \exp \left(- \frac{D}{D_m} \right) \quad (6)$$

The relationship between volume and density is shown in Eq. (7).

$$Q = D \cdot U_f \cdot \exp \left(- \frac{D}{D_m} \right) \quad (7)$$

Meanwhile, the relationship between volume and speed is shown in Eq. (8) [4].

$$Q = U_s \cdot D_m \cdot \ln \left(\frac{U_f}{U_s} \right) \quad (8)$$

The maximum volume (Q_m) can be calculated using Eq. (9) [9].

$$Q_m = \frac{D_m \cdot U_f}{\exp} \quad (9)$$

The graphical relationship between the variables of volume, speed, and density can be depicted using a mathematical equation, which is the basic equation of the traffic flow movement that can be seen in Fig. 1.

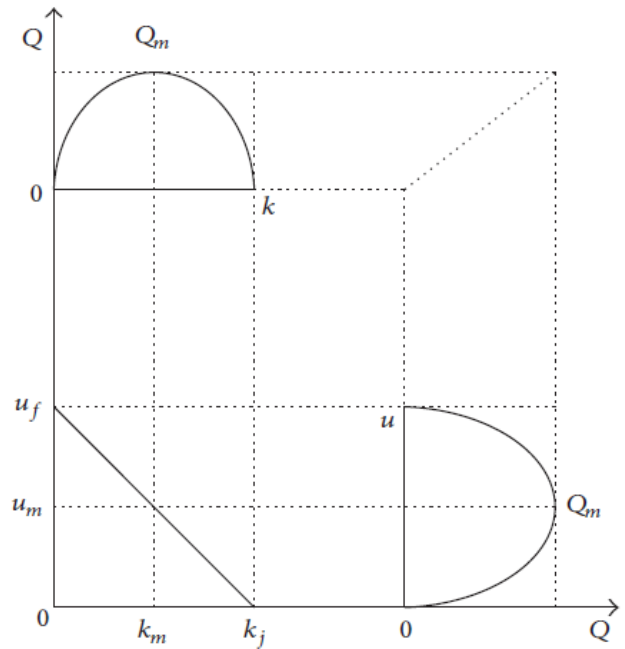


Fig. 1. Graph of volume, speed, and density relationships [11]

This research aims to determine mathematical and graphical relationships between volume, speed, and traffic density with Greenshields and Underwood models on Ir. H. Juanda Street. The objectives to be achieved from the research to be carried out on Ir. H. Juanda Street are to analyze the relationship between volume, speed, and density of traffic with PKJI standard that is suitable for Ir. H. Juanda Street, get the model with the best accuracy, and get the highest maximum volume value between Greenshields and Underwood models.

There is some potential use of Greenshields and Underwood models can be applied in the field to identify

parking space patterns (parallel parking or serial parking) on the shoulder of the road according to road performance [12], [13], identify the width of the road benefit space (sidewalk) as space for pedestrians and non-motorized vehicles [14], and carry out traffic simulations with the program in more depth (microscopic) [15], [16].

2. Method

2.1. Vehicle Classification and Conversion Factor

Vehicles in the traffic flow for PKJI are classified into five categories: Motorcycle (SM), Passenger Car (MP), Medium Vehicle (KS), Big Bus (BB), and Big Truck (TB). The non-motorized vehicle (KTB) type is not converted into traffic flow because it is considered a side barrier.

Passenger car equivalent (EMP) is the conversion factor for medium vehicles, big buses, big trucks, and motorcycles compared to passenger cars in terms of their impact on road capacity. In the capacity analysis, Q must be converted into units of SMP/hour using the EMP value. The EMP value for MP is one [3]. The divided road type can be seen in **Table 1**.

Table 1. EMP for divided road type

Type of Road	Traffic Volume per Route (veh/h)	EMPKS	EMPSM
4/2-T or 2/1	< 1050	1,30	0,40
	≥ 1050	1,20	0,25
6/2-T or 3/1	< 1100	1,30	0,40
	≥ 1100	1,20	0,25

2.2. Linear Regression Analysis and Correlation Coefficient

Linear regression analysis was used to determine the specific characteristics of the traffic flow model, such as speed and density.

If the dependent variable is linear to the independent variable, then the relationship between the two variables is known as linear regression analysis. Multiple linear analysis is called multiple linear analysis if the relationship is more than two independent variables. If the dependent variable (Y) and independent variable (X) have a linear relationship, then the regression function can be seen in **Eq. (10)**.

$$Y = A + BX \quad (10)$$

Measure how far the regression function's accuracy is by looking at the value of the correlation coefficient (r). If r is getting closer to the value of one, then the regression equation is getting better [4].

The correlation coefficient (r) can be used to find the strength of the relationship between variables. The following is an interpretation of the correlation coefficient, which is explained in **Table 2**.

Table 2. Interpretation of correlation coefficient

Interval	Relationship Level
0,00–0,199	Very Low
0,20–0,399	Low

0,40–0,599	Medium
0,60–0,799	Strong
0,80–1,000	Very Strong

The determination coefficient (r^2) [17] can be used to find the effect of the variance of certain variables on other variables. The determination coefficient is calculated by squaring the correlation coefficient found and multiplied by 100%. The determination coefficient is expressed in percent [18].

2.3. Data Collection and Processing

The research location used as a case study is Ir. H. Juanda Street, Central Jakarta. The point used for collecting data on volume and speed is the Harmoni–Pasar Baru road section, as shown in **Fig. 2**.

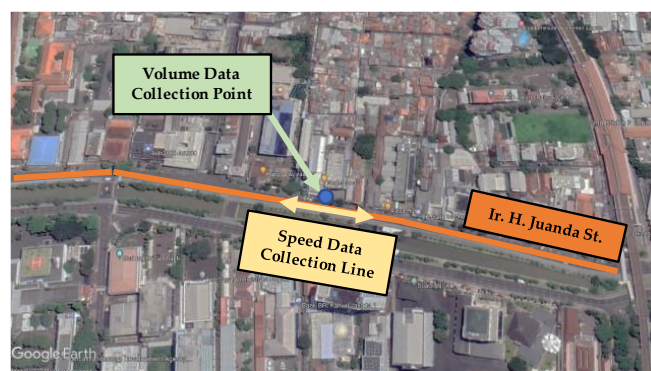


Fig. 2. Research location map

The research will be conducted on Wednesday (June 7, 2023), Friday (March 17, 2023), and Sunday (March 19, 2023), with a data collection time of 11 hours from 07.00 to 18.00. The data collection time is considered according to the level of traffic activity, which is very congested on weekdays and quieter on holidays.

Ir. H. Juanda Street's road type is an 8/2-T or 8-lane 2-way-divided road. This is because the road is divided by Veteran Street through a median in the form of a canal.

Data collection needs materials and tools, as shown in **Table 4**. Techniques for processing primary and secondary data, as explained in **Table 6** and **Table 7**, are also required to provide data processing on specific indicators in **Table 5**.

Table 3. Research targets

No.	Description	Scope	Target
1	Population	Type of Vehicle	Passenger Car (MP)
			Medium Vehicle (KS)
			Big Bus (BB)
			Big Truck (TB)
			Motorcycle (SM)
2	Sample	Volume (Q)	Using the Traffic Counter Application [15-Minute Intervals]

No.	Description	Scope	Target
		Speed (U_s)	Using the Smart Speed Application [Average Speed; 15-Minute Intervals]

As mentioned in **Table 3**, vehicle samples in speed data collection use the recommended sample numbers of at least five vehicles using the local speed method based on 'Guidelines for Surveying and Calculation of Traffic Travel Time No. 001/T/BNKT/1990'. The 50 m road length section will be used to calculate the speed [19].

Table 4. Research materials and tools

No.	Materials and Tools	Role	Total
1	Smartphone	Traffic Counter to Count Volume	2
		Smart Speed to Count Speed	
2	Stationery (Pen)	Note the Results of Observations	2
		Volume Observation Form	
3	Paper	Speed Observation Form	2

Table 5. Research indicators

No.	Indicator	Variable	Equation	Variable Role
1	Speed-Density Relationship [20]	Density (D)	Linear (X)	Independent Variable
		Speed (U_s)	Linear (Y)	Dependent Variable
2	Volume-Density Relationship [21]	Density (D)	Parabolic	Variable Substitution
		Volume (Q)	Parabolic	Variable Substitution
3	Volume-Speed Relationship [22]	Speed (U_s)	Parabolic	Variable Substitution
		Volume (Q)	Parabolic	Variable Substitution

Table 6. Primary data collection techniques

No.	Data	Technique	Data Source
1	Volume (Q)	Site Observation	Ir. H. Juanda Street
2	Speed (U_s)	Site Observation	Ir. H. Juanda Street

Table 7. Secondary data collection techniques

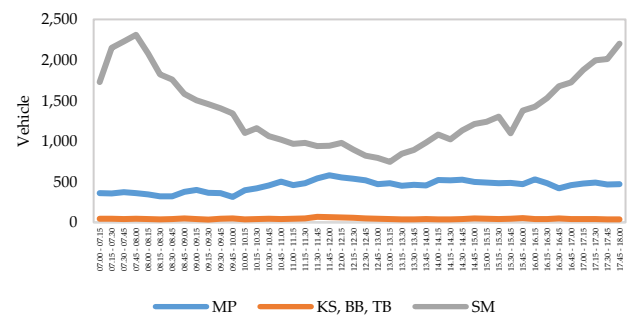
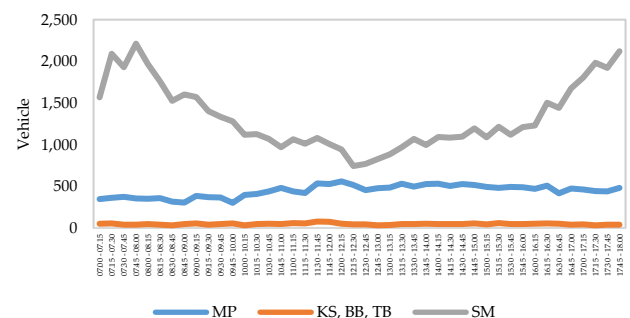
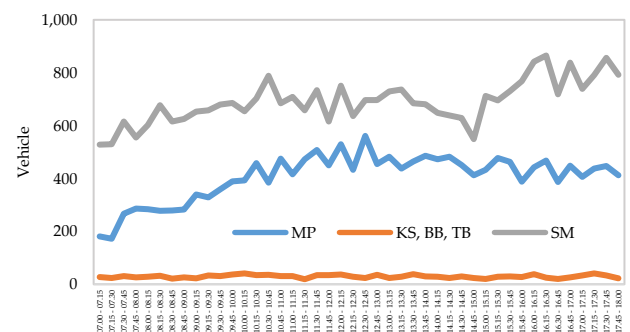
No.	Data	Technique	Data Source
-----	------	-----------	-------------

1	Indonesian Highway Capacity Guidance [23] (PKJI)	Online Source	Directorate General of Highways
2	Road Map	Online Source	Google Earth
3	Journal of 'Hubungan Volume, Kecepatan, dan Kepadatan Lalulintas di Ruas Jalan HR Rasuna Said (Jakarta)'	Online Source	Book of 'Perencanaan dan Pemodelan Transportasi'

3. Result and Discussion

3.1. Basic Relationship Between Volume, Speed, and Density

Primary data, which consists of volume (Q) and speed (U_s), will be processed to obtain density (D). The volume data can be seen in **Fig. 3**, **Fig. 4**, and **Fig. 5**.


Fig. 3. Volume graph on Wednesday

Fig. 4. Volume graph on Friday

Fig. 5. Volume graph on Sunday

Process speed (US) requires speed data from each type of vehicle, which is obtained every 15 minutes (Fig. 6).

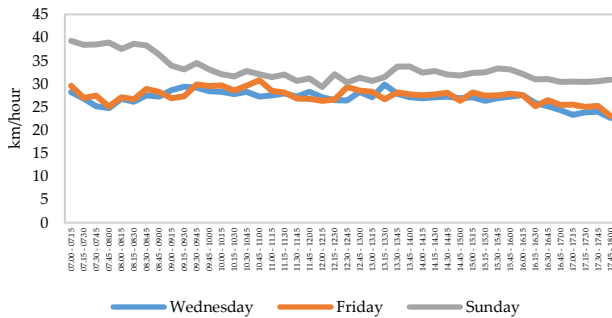


Fig. 6. Speed graph on Wednesday, Friday, and Sunday

To process volume (Q) data as figured in Fig. 7, the data on the number of vehicles obtained every 15 minutes will be multiplied by the passenger car equivalent (EMP) for each type of vehicle based on PKJI guidelines. The multiplication result will be expressed in passenger car units (SMP) [24], [25], [26].

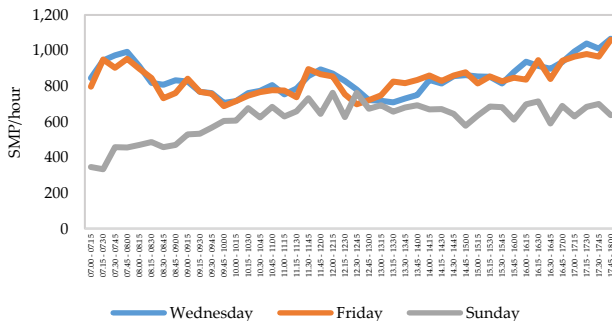


Fig. 7. Volume graph with SMP on Wednesday, Friday, and Sunday

To process density (D) data as figured in Fig. 8, it requires volume (Q) and speed (U_s) data. The volume value (Q) is divided by the speed (U_s) used to find the density number.

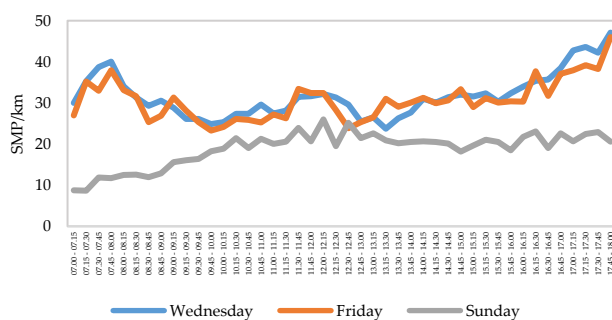


Fig. 8. Density graph on Wednesday, Friday, and Sunday

3.2. Mathematical Relationship with Greenshields Model

The Greenshields model is a linear model that moves continuously without any disturbance. The first step is to find a relationship between speed and density, as shown in Table 11.

To get the constant values U_f and D_j , calculations are carried out with linear regression analysis using the help of the Microsoft Excel 2019 program, as shown in Table 8, Table 9, and Table 10.

Table 8. Greenshields linear regression on Wednesday

Variable	Results
$Y = A + BX$	$Y = 35,505809 - 0,271305 X$
$U_f = A$	$U_f = 35,505809$
$-U_f/D_j = B$	$-35,505809/D_j = -0,271305$
	$D_j = 35,505809/-0,271305$
	$D_j = 130,870470$

Table 9. Greenshields linear regression on Friday

Variable	Results
$Y = A + BX$	$Y = 36,596999 - 0,298119 X$
$U_f = A$	$U_f = 36,596999$
$-U_f/D_j = B$	$-36,596999/D_j = -0,298119$
	$D_j = -36,596999/-0,298119$
	$D_j = 122,759573$

Table 10. Greenshields linear regression on Sunday

Variable	Results
$Y = A + BX$	$Y = 44,700312 - 0,616205 X$
$U_f = A$	$U_f = 44,700312$
$-U_f/D_j = B$	$-44,700312/D_j = -0,616205$
	$D_j = -44,700312/-0,616205$
	$D_j = 72,541304$

Table 11. Mathematical relationship of speed and density with Greenshields

	$U_s = U_f - \left(\frac{U_f}{D_j}\right) \cdot D$
Wednesday	$U_s = 35,505809 - 0,271305 D$
Friday	$U_s = 36,596999 - 0,298119 D$
Sunday	$U_s = 44,700312 - 0,616205 D$

Next, look for the value of the correlation coefficient (r) and the determination coefficient (r^2) from the linear regression analysis with the following results in Table 12.

Table 12. Value of correlation coefficient and determination coefficient with Greenshields

	r = Correlation Coefficient and r^2 = Determination Coefficient
Wednesday	$r = -0,922628$ (Very Strong (-) Decrease Occurred) $r^2 = 0,851243$
Friday	$r = -0,932503$ (Very Strong (-) Decrease Occurred) $r^2 = 0,869561$
Sunday	$r = -0,940426$ (Very Strong (-) Decrease Occurred) $r^2 = 0,884401$

In the Greenshield model, the correlation values on Wednesday, Friday, and Sunday have values > 90 so that they are included in the very strong category so that the relationship between the independent variable (X) on density (D) and the dependent variable (Y) on speed (U_s) is very strong. The resulting curve line will decrease because the value of the correlation coefficient is negative.

In terms of the volume and speed relationship with the Greenshields model, the equation used is parabolic, with the following results in **Table 13**.

Table 13. Mathematical relationship of volume and speed with Greenshields

	$Q = D_j \cdot U_s - \left(\frac{D_j}{U_f}\right) \cdot U_s^2$
Wednesday	$Q = 130,870470 U_s - 3,685889 U_s^2$
Friday	$Q = 122,759573 U_s - 3,354362 U_s^2$
Sunday	$Q = 72,541304 U_s - 1,622837 U_s^2$

The relationship between volume and density with the Greenshields model is also parabolic, with the following results in

Table 14.

Table 14. Mathematical relationship of volume and density with Greenshields

	$Q = U_f \cdot D - \left(\frac{U_f}{D_j}\right) \cdot D^2$
Wednesday	$Q = 35,505809 D - 0,271305 D^2$
Friday	$Q = 36,596999 D - 0,298119 D^2$
Sunday	$Q = 44,700312 D - 0,616205 D^2$

After obtaining the mathematical equations of the three relationships, the maximum volume can be found in **Table 15**.

Table 15. Maximum volume with Greenshields

	$Q_m = \frac{D_j \cdot U_f}{4}$
Wednesday	$Q_m = \frac{130,87047 \times 35,505809}{4}$ = 1.161,665481 SMP/hour
Friday	$Q_m = \frac{122,759573 \times 36,596999}{4}$ = 1.123,157982 SMP/hour
Sunday	$Q_m = \frac{72,541304 \times 44,700312}{4}$ = 810,654739 SMP/hour

3.3. Mathematical Relationship with the Underwood Model

The Underwood model is an exponential model with different equations from the Greenshields model. This difference can be seen from the density constant used in the Greenshields model, which is the density during traffic jams or jam density (D_j). In contrast, the Underwood model is the density during maximum volume (D_m).

To get the constant values U_f and D_m , calculations are

carried out with linear regression analysis using the help of the Microsoft Excel 2019 program, as shown in **Table 16**, **Table 17**, and **Table 18**.

Table 16. Underwood linear regression on Wednesday

Variable	Results
$Y = A + BX$	$Y = 3,620994 - 0,010424 X$
$\ln U_f = A$	$\ln U_f = 3,620994$ $U_f = 37,374683$
$-1/D_m = B$	$-1/D_m = -0,010424$ $D_m = 95,936923$

Table 17. Underwood linear regression on Friday

Variable	Results
$Y = A + BX$	$Y = 3,650702 - 0,011079 X$
$\ln U_f = A$	$\ln U_f = 3,6502702$ $U_f = 38,501702$
$-1/D_m = B$	$-1/D_m = -0,011079$ $D_m = -1/-0,011079$ $D_m = 90,260862$

Table 18. Underwood linear regression on Sunday

Variable	Results
$Y = A + BX$	$Y = 3,833421 - 0,017908 X$
$\ln U_f = A$	$\ln U_f = 3,833421$ $U_f = 46,220375$
$-1/D_m = B$	$-1/D_m = -0,017908$ $D_m = -1/-0,017908$ $D_m = 55,841001$

From the results that have been obtained above, the regression equation for the relationship between speed and density can be seen in **Table 19**.

Table 19. Mathematical relationship of speed and density with Underwood

	$U_s = U_f \cdot \exp\left(-\frac{D}{D_m}\right)$
Wednesday	$U_s = 37,374683 \exp(-0,010424 D)$
Friday	$U_s = 38,501702 \exp(-0,011079 D)$
Sunday	$U_s = 46,220375 \exp(-0,017908 D)$

Next, look for the value of the correlation coefficient (r) and the determination coefficient (r^2) from the linear regression analysis with the following results in **Table 20**.

Table 20. Value of correlation coefficient and determination coefficient with Underwood

	r = Correlation Coefficient and r^2 = Determination Coefficient
Wednesday	$r = -0,926928$ (Very Strong (-) Decrease Occurred) $r^2 = 0,859196$

Friday	$r = -0,938553$ (Very Strong (-) Decrease Occurred) $r^2 = 0,880881$
Sunday	$r = -0,940745$ (Very Strong (-) Decrease Occurred) $r^2 = 0,885001$

In the Underwood model, the correlation values on Wednesday, Friday, and Sunday have values > 90 so that they are included in the very strong category that resulted in a relationship between the independent variable (X) on density (D) and the dependent variable (Y) on speed (U_s) being very strong. The resulting curve line will decrease because the value of the correlation coefficient is negative.

The relationship between volume and speed in the Underwood model is parabolic so that it can be seen in **Table 21**.

Table 21. Mathematical relationship of volume and speed with Underwood

	$Q = U_s \cdot D_m \cdot \ln \left(\frac{U_f}{U_s} \right)$
Wednesday	$Q = 95,936923 U_s \ln \left(\frac{37,374683}{U_s} \right)$
Friday	$Q = 90,260862 U_s \ln \left(\frac{38,501702}{U_s} \right)$
Sunday	$Q = 55,841001 U_s \ln \left(\frac{46,220375}{U_s} \right)$

After obtaining the volume and speed relationship, the volume and density relationship can be seen in **Table 22**.

Table 22. Mathematical relationship of volume and density with Underwood

	$Q = D \cdot U_f \cdot \exp \left(-\frac{D}{D_m} \right)$
Wednesday	$Q = 37,374683 D \exp (-0,010424 D)$
Friday	$Q = 38,501702 D \exp (-0,011079 D)$
Sunday	$Q = 46,220375 D \exp (-0,017908 D)$

The maximum volume in the Underwood model can be seen in **Table 23**.

Table 23. Maximum volume with Underwood

	$Q_m = \frac{D_m \cdot U_f}{\exp}$
Wednesday	$Q_m = \frac{95,936923 \times 37,374683}{\exp}$ $= 1.319,07296 \text{ SMP/hour}$
Friday	$Q_m = \frac{90,260862 \times 38,501702}{\exp}$ $= 1.278,453481 \text{ SMP/hour}$

$$\text{Sunday} \quad Q_m = \frac{55,841001 \times 46,220375}{\exp} = 949,493903 \text{ SMP/hour}$$

The maximum volume (Q_m) on the Greenshields and Underwood models is similar to the road capacity (C) on PKJL. Road capacity and maximum volume use free-flow speed and capacity (volume) parameters as important parameters in research to determine driver behavior and vehicle population through field measurements.

The Greenshields model has similarities with the Greenberg model, which requires parameters such as D_j and U_m . The U_m parameter in the Greenshields model can be calculated as half of the U_f value. Meanwhile, the Greenberg model cannot calculate the free-flow speed (U_f). This makes the jam density (D_j) more difficult to observe in the field and the maximum speed (U_m) more difficult to estimate than the free-flow speed (U_f).

The Underwood model has similarities with the Greenshields model, which requires U_f and D_m parameters. The D_m parameter in the Greenshields model can be calculated as half of the D_j value. Free-flow speed is easier to estimate, so there is no need to look for the jam density, which has a value that is approximately the same as the maximum density.

3.4. Graphical Relationship with Greenshields and Underwood Models

The previously obtained mathematical relationship between volume, speed, and density will produce a suitable curve shape through a graphic image using the Greenshields and Underwood models in **Fig. 9**, **Fig. 10**, and **Fig. 11**.

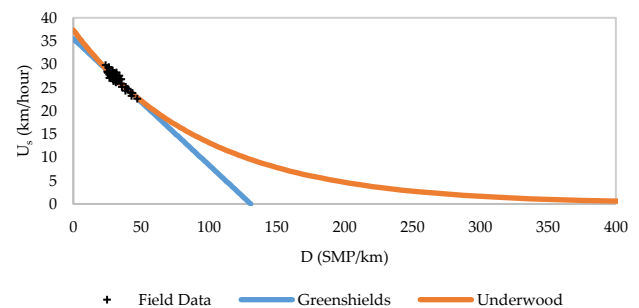


Fig. 9. Graphical relationship of speed and density with Greenshields and Underwood models on Wednesday

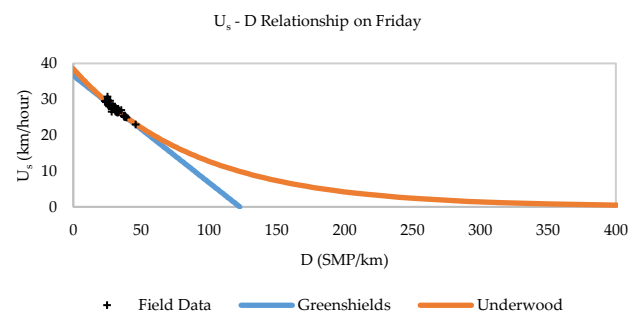


Fig. 10. Graphical relationship of speed and density with Greenshields and Underwood models on Friday

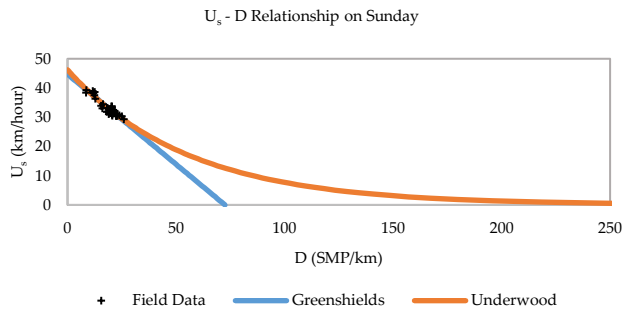


Fig. 11. Graphical relationship of speed and density with Greenshields and Underwood models on Sunday

Graphical depictions of the relationship between speed (U_s) and density (D) on Wednesday, Friday, and Sunday by **Fig. 9**, **Fig. 10**, and **Fig. 11** are made with the X-axis acting as density (D) and the Y-axis acting as speed (U_s). Based on the Greenshields (in blue) and Underwood (in orange) models, the curve line tends to decrease when the density increases and the speed decreases. When the speed point is zero, the jam density will be reached, and congestion will occur. When the density point is zero, the free-flow speed will be reached, and the traffic conditions will be smooth.

The volume indicator on **Fig. 9**, **Fig. 10**, and **Fig. 11** is not visible when determining the maximum volume, but the maximum volume can be achieved when the linear line is at the halfway or midpoint of a line. As a result, the largest maximum volume was obtained by the Underwood model at 1.319,07296 SMP/hour on Wednesday with coordinates D and U_s at (95,93692; 13,74938).

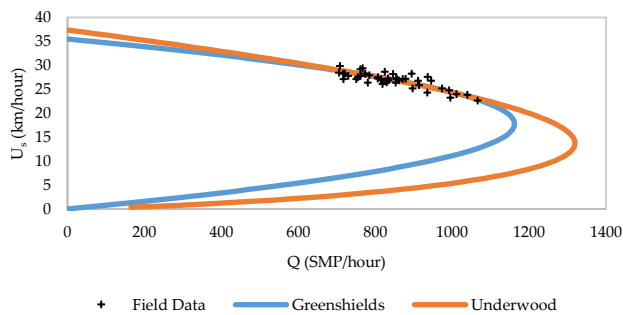


Fig. 12. Graphical relationship of volume and speed with Greenshields and Underwood models on Wednesday

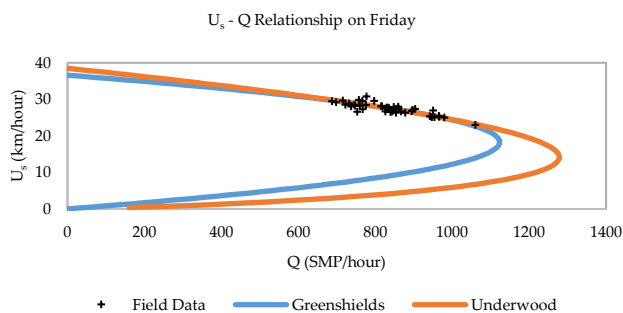


Fig. 13. Graphical relationship of volume and speed with Greenshields and Underwood models on Friday

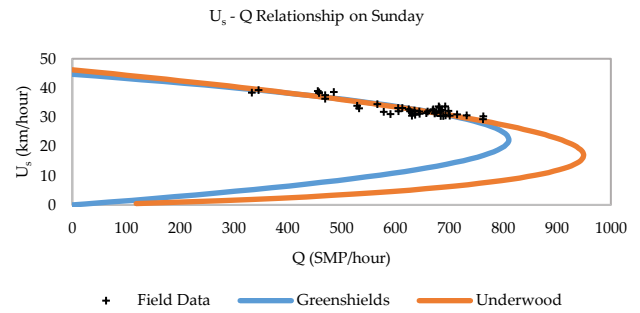


Fig. 14. Graphical relationship of volume and speed with Greenshields and Underwood models on Sunday

Graphical depictions of the relationship between volume (Q) and speed (U_s) on Wednesday, Friday, and Sunday in **Fig. 12**, **Fig. 13**, and **Fig. 14** are made with the X-axis acting as volume (Q) and the Y-axis acting as speed (U_s). The curved lines based on the Greenshields (in blue) and Underwood (in orange) models will form a parabola horizontally and face to the right. With each increase in volume up to the maximum point (maximum volume), the speed decreases. Congestion decreases the volume and speed after the maximum volume is reached.

The volume indicator on **Fig. 12**, **Fig. 13**, and **Fig. 14** is visible when determining the maximum volume. So, the maximum volume can be achieved when the horizontal path of the parabola facing right experiences a sharp turn facing left so that it produces a peak point. As a result, the largest maximum volume was obtained by the Underwood model at 1.319,07296 SMP/hour on Wednesday with coordinates Q and U_s at (1.319,07296; 13,74938).

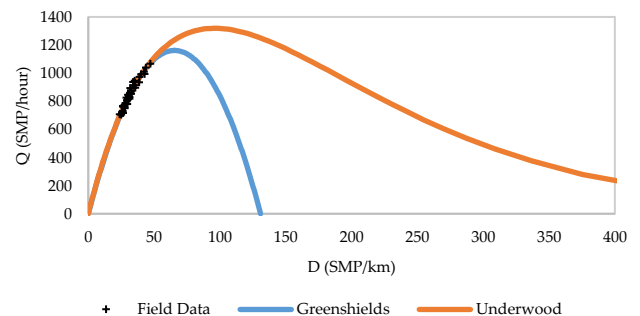


Fig. 15. Graphical relationship of volume and density with Greenshields and Underwood models on Wednesday

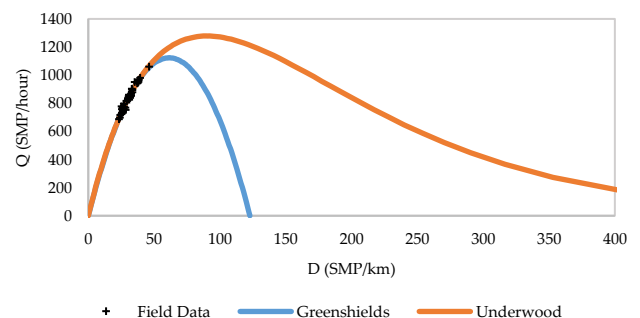


Fig. 16. Graphical relationship of volume and density with Greenshields and Underwood models on Friday

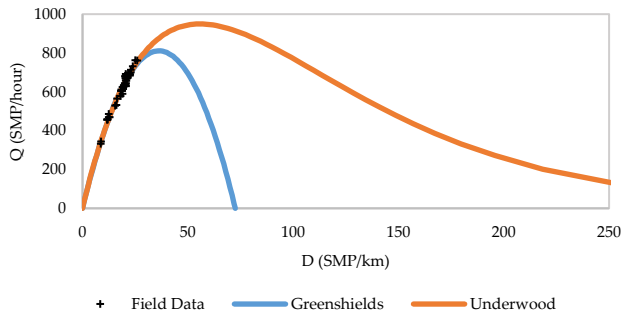


Fig. 17. Graphical relationship of volume and density with Greenshields and underwood models on Sunday

Graphical depictions of the relationship between volume (Q) and density (D) on Wednesday, Friday, and Sunday by **Fig. 15**, **Fig. 16**, and **Fig. 17** are made with the X-axis acting as density (D) and the Y-axis acting as volume (Q). The curved line based on the Greenshields (in blue) and Underwood (in orange) models will form a parabola vertically and face upward. With each increase in volume up to the maximum point (maximum volume), the density will reach the maximum point (maximum density) so that the vehicle capacity in one road section will be fulfilled. After the volume decreases, the density will continue to increase until congestion occurs.

The volume indicator on **Fig. 15**, **Fig. 16**, and **Fig. 17** is visible when determining the maximum volume. So, the maximum volume can be achieved when the vertical path of the parabola facing upward experiences a sharp turn facing downward so that it produces a peak point. As a result, the largest maximum volume was obtained by the Underwood model at 1.319,07296 SMP/hour on Wednesday with coordinates D and Q at (95,93692; 1.319,07296).

The analysis above shows that the growth of traffic flow in the Underwood model is more significant than in the Greenshields model. The traffic flow will start with a high speed according to the wishes of the driver, which is called the free-flow speed (U_f); after the free-flow speed is reached when $Q = 0$ and $D = 0$, the vehicle will continue to increase, and the maximum volume (Q_m) will occur. The maximum volume will have a curved line intersecting the maximum density (D_m) and maximum speed (U_m). After the maximum volume is reached, the vehicles will pile up, and jam density (D_j) will occur so that the curve line will not intersect with the maximum volume, and the space mean speed (U_s) and volume (Q) will decrease.

4. Conclusion

Based on the analysis results, the relationship between volume, speed, and traffic density with PKJI standard is suitable for Ir. H. Juanda Street on the Underwood model. This is because the r value of 0,940475 shows the strongest relationship between speed and density to produce an equation $U_s = 46,22 \exp(-0,02 D)$, $Q = 55,84 U_s \ln(46,22/U_s)$, and $Q = 46,22 D \exp(-0,02 D)$. The best accuracy model is obtained using the Underwood model on Sunday with the r^2 value of 0,885001. The determination coefficient shows how

well the regression prediction of the model is based on field data with an accuracy of 88,5%. The highest maximum volume was obtained using the Underwood model on Wednesday at 1.319,07296 SMP/hour.

Author Declaration

Authors' contributions and responsibilities

The authors made substantial contributions to the conception and design of the study. The authors took responsibility for data analysis, interpretation, and discussion of results. The authors read and approved the final manuscript.

Availability of data and materials

All data are available from the authors.

Competing interests

The authors declare no competing interest.

References

- [1] Kementerian PUPR, Peraturan Menteri Pekerjaan Umum dan Perumahan Rakyat Nomor 05/PRT/M/2018 tentang Penetapan Kelas Jalan Berdasarkan Fungsi dan Intensitas Lalu Lintas serta Daya Dukung Menerima Muatan Sumbu Terberat dan Dimensi Kendaraan Bermotor (in Bahasa), Jakarta, 2018.
- [2] N. J. Garber and L. A. Hoel, Traffic and Highway Engineering, Enhanced 5th Edition. Canada: Cengage Learning, 2019.
- [3] Direktorat Jenderal Bina Marga, Pedoman Kapasitas Jalan Indonesia (in Bahasa), Jakarta, 2023.
- [4] O. Z. Tamin, "Hubungan Volume, Kecepatan dan Kepadatan Lalu Lintas di Ruas Jalan HR Rasuna Said (Jakarta) (in Bahasa)," *Jurnal Teknik Sipil*, vol. 5, 1992.
- [5] O. Z. Tamin, Perencanaan dan Pemodelan Transportasi, Edisi Kedua (in Bahasa). Bandung: ITB, 2000.
- [6] B. D. Greenshields, "A Study of Traffic Capacity," *Highway Research Board Proceedings*, 1935, pp. 448–477.
- [7] R. Kucharski and A. Drabicki, "Estimating Macroscopic Volume Delay Functions with the Traffic Density Derived from Measured Speeds and Flows," *J Adv Transp*, vol. 2017, pp. 1–10, 2017, doi: 10.1155/2017/4629792.
- [8] C. J. Khisty and B. K. Lall, Dasar-dasar Rekayasa Transportasi Jilid 1 Edisi Ketiga (in Bahasa), Jakarta: Erlangga, 2006.
- [9] B. Saputra and D. Savitri, "Analisis Hubungan antara Volume, Kecepatan dan Kepadatan Lalu-Lintas Berdasarkan Model Greenshield, Greenberg dan Underwood (in Bahasa)," *Jurnal Manajemen Aset Infrastruktur & Fasilitas*, vol. 5, no. 1, 2021.
- [10] R. T. Underwood, "Speed, Volume, and Density Relationships," *Quality and Theory of Traffic Flow*, New Haven: Bureau of Highway Traffic, Yale University, 1961, pp. 141–188.
- [11] C. Yu, J. Zhang, D. Yao, R. Zhang, and H. Jin, "Speed-Density Model of Interrupted Traffic Flow Based on Coil Data," *Mobile Information Systems*, vol. 2016, pp. 1–12,

- 2016, doi: 10.1155/2016/7968108.
- [12] M. Akbar, J. Paresa, and D. L. Pamuttu, "Analysis of the Effect of Parking on Road Bodies on Road Service Levels," *IOP Conf Ser Mater Sci Eng*, vol. 1125, no. 1, p. 012014, May 2021, doi: 10.1088/1757-899X/1125/1/012014.
- [13] J. Parmar, P. Das, and S. M. Dave, "Study on demand and characteristics of parking system in urban areas: A review," *Journal of Traffic and Transportation Engineering (English Edition)*, vol. 7, no. 1, pp. 111–124, Feb. 2020, doi: 10.1016/j.jtte.2019.09.003.
- [14] N. A. Coppola and W. E. Marshall, "Sidewalk Static Obstructions and Their Impact on Clear Width," *Transp Res Rec*, vol. 2675, no. 6, pp. 200–212, 2021, doi: 10.1177/0361198121991833.
- [15] P. Sun, X. Wang, and M. Zhu, "Modeling Car-Following Behavior on Freeways Considering Driving Style," *J Transp Eng A Syst*, vol. 147, no. 12, p. 4021083, 2021, doi: 10.1061/JTEPBS.0000584.
- [16] M. Seraj, J. Li, and Z. Qiu, "Modeling Microscopic Car-Following Strategy of Mixed Traffic to Identify Optimal Platoon Configurations for Multiobjective Decision-Making," *J Adv Transp*, vol. 2018, pp. 1–15, Sep. 2018, doi: 10.1155/2018/7835010.
- [17] R. Sun, C. Gao, and M. Zhang, "Study on Speed-Flow Relationship Model of Urban Roadways," *CICTP 2020*, 2020, pp. 3482–3489. doi: 10.1061/9780784482933.300.
- [18] Sugiyono, *Metode Penelitian Kuantitatif, Kualitatif, dan R&D* 19th ed. (in Bahasa), Bandung: Alfabeta, 2013.
- [19] Direktorat Pembinaan Jalan Kota, *Panduan Survei dan Perhitungan Waktu Perjalanan Lalu Lintas (in Bahasa)*, Jakarta, 1990.
- [20] G. Ren and S. Zhu, "Parameter Calibration of Traffic Flow Speed-Density Model Based on K-means Clustering Algorithm and Least Square Method," *CICTP 2021*, 2021, pp. 66–76. doi: 10.1061/9780784483565.007.
- [21] J. M. del Castillo, "Three New Models for The Flow-Density Relationship: Derivation and Testing for Freeway and Urban Data," *Transportmetrica*, vol. 8, no. 6, pp. 443–465, Nov. 2012, doi: 10.1080/18128602.2011.556680.
- [22] Y. Chen, "Model of Traffic Speed-Flow Relationship at Signal Intersections," *Open Journal of Applied Sciences*, vol. 07, no. 06, pp. 319–327, 2017, doi: 10.4236/ojapps.2017.76025.
- [23] A. K. Indriastuti and E. Yulipriyono, "Indonesian Highway Capacity Guidance (IHCG) Correction Factor for Signalized Intersection," *MATEC Web of Conferences*, vol. 159, p. 01024, Apr. 2018, doi: 10.1051/mateconf/201815901024.
- [24] Direktorat Jenderal Bina Marga, *Manual Kapasitas Jalan Indonesia (in Bahasa)*, Jakarta, 1997.
- [25] Transportation Research Board, *Highway Capacity Manual*. Washington, DC: National Academy of Sciences, 2000.
- [26] Transportation Research Board, *Highway Capacity Manual*, Washington, DC: National Academy of Sciences, 2010.